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Computer Science and Artificial Intelligence

Darp: AI-Based Early Risk Detection in Crowded and High-Temperature Environments

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This work is dedicated to my dear parents, the most loving in this world.

ABSTRACT

Large-scale mass gatherings such as the Hajj season present significant health and safety challenges due to extreme crowd density and elevated temperatures, increasing the risks of heat exhaustion and fainting. Current solutions often address crowd management and health monitoring separately, creating a gap in integrated real-time risk prevention.

This project proposes a lightweight AI-based prototype for early risk detection in crowded and high-temperature environments. The system employs a Convolutional Neural Network (CNN), trained on the ShanghaiTech dataset, to classify crowd density with a target accuracy of at least 85%. Crowd density is integrated with normalized temperature values and user-reported symptoms through a rule-based engine to compute a weighted baseline risk score.

The system generates location-based alerts to support medical teams in prioritizing rapid intervention in high-risk zones. It also provides preventive notifications for pilgrims within the same area without offering medical diagnosis or storing personal data.

The expected outcome is a fully functional web-based prototype that enables spatial risk monitoring and supports timely decision-making while preserving privacy.

Keywords: Hajj Safety, Crowd Density Estimation, Heat Risk Assessment, Early Warning System.

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CHAPTER 1

INTRODUCTION AND MOTIVATION

1.1 Problem Context and Application Domain

This project falls within the field of Artificial Intelligence, with a focus on Computer Vision and its applications in risk management within crowded environments. The system targets large-scale mass gatherings such as the Hajj season, where high crowd density intersects with extreme environmental conditions.

The core problem lies in the increased likelihood of sudden health risks, such as heat exhaustion and fainting, under these conditions. Detecting early warning signs and responding promptly becomes challenging due to the dynamic and highly congested nature of such environments.

Although existing solutions provide crowd monitoring or health tracking, they operate independently, resulting in a lack of a comprehensive, real-time understanding of the overall risk level.

1.2 Motivation and Significance

The motivation behind this project stems from the need to overcome the limitations of current systems that rely on partial analysis of the problem.

Effective risk assessment in crowded environments requires the integration of multiple interrelated factors, including crowd density, environmental conditions, and basic health indicators. Relying on a single factor leads to incomplete evaluation and delayed response.

This project is significant as it introduces an integrated approach that combines these factors within a unified framework, enhancing early risk detection and enabling more proactive and informed decision-making in high-risk environments.

1.3 High-Level Project Vision

The project aims to develop a lightweight intelligent system capable of performing real-time risk assessment by integrating multiple data sources.

The system utilizes Computer Vision techniques to estimate crowd density, alongside environmental temperature data and simple user-reported health indicators. By combining these inputs, the system generates an accurate risk level assessment for each monitored area.

The system is designed to support authorities and medical teams through continuous monitoring and real-time alerts, while also allowing users to interact by reporting simple symptoms and receiving preventive notifications.

1.4 Expected Academic or Practical Impact

The expected outcome of this project is the development of an intelligent system capable of dynamically classifying risk levels into clear categories, enabling faster and more effective decision-making.

Practically, the system enhances response efficiency by identifying high-risk areas and generating early alerts, which supports better resource allocation and reduces the likelihood of incidents.
Academically, the project demonstrates the value of integrating multiple data sources within a unified AI-based framework and highlights the impact of such integration on improving real-time risk assessment in complex environments.

CHAPTER 2

BACKGROUND AND RELATED WORK

Literature Selection Criteria

The studies included in this review were selected based on a set of well-defined criteria to ensure relevance, quality, and reliability. Priority was given to recent publications from 2020 onwards in order to reflect the latest advancements in Artificial Intelligence and related technologies. The selected studies are directly related to key topics of this project, including crowd analysis, heat-related risk assessment, and AI-based healthcare monitoring systems.In addition, only peer-reviewed journal articles and conference papers from reputable publishers such as IEEE, ACM, Elsevier, and MDPI were considered to ensure academic credibility. Studies that provide experimental validation, measurable performance results, or real-world applications were prioritized over purely theoretical work.Furthermore, diversity in the selected literature was ensured to cover different aspects of the problem, including crowd monitoring systems, digital health solutions, and environmental risk analysis. The inclusion of modern techniques such as deep learning, convolutional neural networks (CNNs), transformer-based models, and Internet of Things (IoT) technologies was also considered.Finally, only accessible and verifiable sources available through platforms such as Google Scholar and open-access repositories were included, ensuring transparency and the ability to validate all referenced studies.

2.1 Relevant Concepts and Technologies

Recent advancements in Artificial Intelligence (AI) have significantly enhanced the ability to analyze complex and dynamic environments such as mass gatherings. In particular, Computer Vision and Deep Learning, especially Convolutional Neural Networks (CNNs), have become the core technologies for crowd analysis and scene understanding.

Crowd analysis has evolved from traditional image processing techniques to deep learning–based models capable of estimating density and detecting congestion in real time. Recent surveys highlight that CNN-based methods significantly improve accuracy in crowd density estimation, especially in dense and occluded environments [12], [13]. Transformer-based architectures have also been introduced to improve feature extraction in highly congested scenes [14].

In parallel, health monitoring systems have gained attention with the integration of Artificial Intelligence and Internet of Things (IoT) technologies. These systems allow continuous monitoring of health-related indicators and support early detection of potential risks. Recent studies show that AI-based healthcare systems can process real-time data and improve decision-making in emergency situations [16], [17].

Another critical concept is heat-related risk modeling, which is especially important in extreme environments such as Hajj. Research indicates that prolonged exposure to high temperatures significantly increases the risk of heat stroke and heat exhaustion, particularly in crowded environments where mobility is limited [19]. The World Health Organization also emphasizes the importance of preventive monitoring systems during mass gatherings to reduce heat-related health risks [18].

2.2 Existing Systems and Solutions

Existing systems in the literature address crowd monitoring, digital health, and environmental risk analysis, yet they differ in scope, performance, and level of integration. Digital health systems such as [1] and [2] focus on symptom-checking and user interaction, showing positive results in improving accessibility and response time; however, they rely heavily on user input and lack environmental awareness, making them only partially suitable for large-scale environments. Crowd monitoring approaches in [3] and [4] show that computer vision and deep learning techniques can effectively estimate crowd density and analyze movement patterns, achieving high accuracy in complex scenes, although they are limited to visual data and do not consider health-related risks. Medical and heat-related studies such as [5] and [6] highlight the serious impact of high temperatures on human health, confirming that heat exhaustion and heat stroke are critical risks; however, these studies are mostly clinical and reactive rather than real-time or predictive. Similarly, [7] and [8] present CNN-based models for crowd density estimation with strong performance in dense environments, but they require large datasets and struggle in highly occluded scenes. Environmental studies such as [9] further emphasize the role of heat as a major risk factor, although they lack integration with intelligent systems. Composite modeling approaches like [10] provide a useful framework for combining multiple risk factors, offering flexibility in weighting different variables, but they depend on predefined parameters. Dataset-oriented research such as [11] contributes standardized benchmarks for training and evaluation yet lacks contextual information such as environmental conditions. Survey-based studies in [12] and [13] confirm the superiority of deep learning methods for crowd analysis, although they do not propose practical implementations. More advanced approaches such as the transformer-based model in [14] improve feature extraction and global context understanding in dense crowds, but introduce high computational complexity, while scale-adaptive models in [15] enhance performance across varying densities, despite increased training complexity. Healthcare monitoring systems in [16] and [17] demonstrate that integrating AI and IoT enables continuous monitoring and improves decision-making; however, these systems depend on wearable devices and do not incorporate environmental or crowd-related factors. Preventive health guidelines such as [18] emphasise the importance of proactive monitoring in mass gatherings, although they remain non-technical. Clinical studies such as [19] confirm that early detection of heat stroke is essential to reduce mortality, but lack real-time implementation mechanisms, while [20] shows that the combination of heat and crowd density significantly increases risks in Hajj environments. Real-time crowd monitoring systems such as [21] achieve high accuracy in dynamic environments, although they depend on camera quality and infrastructure, and density estimation models such as [22] perform well but face challenges in extreme congestion. Smart system approaches in [23] demonstrate that AI-based early warning systems can improve response time, yet they lack full integration across domains, while [24] presents an IoT-based Hajj monitoring system with promising tracking capabilities but without real-time crowd density fusion. Finally, context-aware systems in [25] show that integrating multiple data sources improves adaptability and system intelligence, although they increase system complexity and implementation cost. Overall, these studies show that while most approaches achieve positive results within their specific domains, they remain limited by a lack of integration, indicating that none of the existing solutions fully address the combined challenges of crowd density, heat stress, and health monitoring in a unified real-time system, which justifies the need for the proposed approach

2.3 Comparative Analysis

Number and title and of reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference Between Darb and Other Systems
[1] Digital Symptom Checker System	2023	Rule-based AI, NLP	Fast symptom analysis	Depends on user input, no environmental awareness	Quick response	Integrates symptoms with crowd and temperature
[2] AI-based Health Diagnosis	2024	Machine Learning, NLP	Interactive diagnosis	Not real-time, user-dependent	Moderate accuracy	Adds real-time environmental and crowd data
[3] CNN Crowd Density Estimation	2023	CNN, Computer Vision	High accuracy in dense scenes	Ignores health risks	Accurate detection	Combines density with health and temperature
[4] Real-Time Crowd Monitoring	2024	DeepLearning, Video Analysis	Real-time monitoring	Depends on camera quality	Effective performance	Adds health risk predictio
[5] Heat Stroke Clinical Study	2022	Medical Analysis	Reliable insights	Not real-time	Accurate results	Converts to predictive syste
[6] Heat Stress Risk Modeling	2023	Statistical Models	Identifies heat risks	No AI integration	Accurate risk detection	Integrates heat with AI syste
[7] Crowd Counting using CNN	2023	CNN	Works well in dense crowds	Requires large datasets	High performance	Uses efficient real-time classificatio
[8] Deep Learning Crowd Classification	2024	Deep Learning	Good classification	Struggles in extreme congestion	Moderate performance	Uses multi-factor analysi
[9] Environmental Heat Analysis	2022	Data Analysis	No intelligent integration	No intelligent integration	Useful insights	Integrates environment with AI
[10] Multi-factor Risk Model	2023	ML Models	Combines multiple risks	Static weights	Flexible modeling	Uses adaptive dynamic weights
[11] Crowd Dataset Benchmark	2021	Dataset Processing	Standardized evaluation	Lacks real-world context	Useful training data	Uses real-time contextual data
[12] Survey on Crowd Analysis	2023	Survey	Comprehensive review	No implementation	Analytical insights	Provides practical system

[13] Deep Learning Crowd Survey	2024	Survey, DL	High accuracy comparison	No real system	Evaluation only	Applies in real environment
[14] Transformer Crowd Model	2023	Transformer	Strong feature extraction	High complexity	High accuracy	Uses lightweight approac
[15] Scale-adaptive Crowd Model	2024	Deep Learning	Handles varying densities	Complex training	Improved adaptability	Simplified for real-tim
[16] AI Healthcare Monitoring	2023	AI, IoT	Continuous monitoring	Requires wearable devices	Real-time tracking	No wearable dependenc
[17] Smart Health IoT System	2024	IoT Sensors	Accurate tracking	Limited scalability	Good monitoring	Uses non-intrusive method
[18] WHO Heat Guidelines	2022	Guidelines	Preventive awareness	Not technical	Risk reduction	Converts to automated syste
[19] Heat Stroke Detection Study	2023	Medical Models	Early detection	No real-time system	High accuracy	Enables real-time alert
[20] Hajj Risk Study	2024	Data Analysis	Shows combined risks	No solution provided	Risk identification	Provides integrated solutio
[21] Real-Time Crowd System	2023	AI, CV	Real-time monitoring	Infrastructure dependent	High accuracy	Adds multi-factor analysi
[22] Density Estimation Model	2022	ML	Good estimation	Weak in extreme congestion	Moderate accuracy	Improves robustnes
[23] AI Early Warning System	2024	AI, Rule-based	Fast alerts	Limited integration	Quick response	Uses richer data source
[24] IoT Hajj Monitoring System	2023	IoT	Tracks individuals	No full integration	Good tracking	Integrates crowd + health + environmen
[25] Context-Aware AI System	2024	AI, Context-aware	Adaptive decisions	High complexity	High performance	Balances efficiency and accurac

Table 1: Comparative Analysis

2.4 Identified Gaps and Limitations

Reference	Approach Type	Identified Gaps and Limitations
[1], [2]	Digital Health Systems	Rely heavily on user input, lack environmental awareness (temperature/crowd), not suitable for large-scale crowded environments
[3], [4]	Crowd Monitoring (CV/DL)	Focus only on visual data, do not link crowd density to health risks
[5], [6]	Medical / Heat Studies	Clinical and reactive, not designed for real-time or predictive applications
[7], [8]	CNN-based Density Models	Require large datasets, struggle with occlusion in dense scenes
[9]	Environmental Studies	Focus only on temperature, lack integration with intelligent systems
[10]	Composite Models	Depend on predefined/static weights, limited adaptability
[11]	Dataset-based Research	Provide standardized datasets but lack contextual/environmental information
[12], [13]	Survey Studies	Theoretical analysis only, no practical implementation
[14]	Transformer-based Models	High computational complexity, difficult for real-time deployment
[15]	Scale-adaptive Models	Complex training process, require careful tuning
[16], [17]	AI + IoT Healthcare Systems	Depend on wearable devices, do not consider crowd or environmental factors
[18]	Preventive Guidelines	Non-technical, lack intelligent system implementation
[19]	Clinical Heat Stroke Studies	No real-time implementation mechanisms
[20]	Hajj Risk Studies	Highlight the problem only, no integrated technical solution
[21]	Real-time Crowd Systems	Depend on camera quality and infrastructure
[22]	Density Estimation Models	Face challenges in extremely congested environments
[23]	Smart Early Warning Systems	Lack full integration across domains (health, crowd, environment)
[24]	IoT-based Hajj System	Do not integrate crowd density with health and environmental data
[25]	Context-aware Systems	High system complexity and implementation cost

Table 2: Identified gap and limitations

Key Insight (Research Gap):
Most existing approaches focus on a single aspect (crowd, health, or environment) and lack real-time integration.
No system fully combines crowd density, temperature, and health monitoring in a unified real-time framework.

2.5 Project Contribution

To address these limitations, this project proposes an integrated AI-based system that combines:

- CNN-based crowd density estimation for real-time spatial analysis
- Heat risk modeling based on environmental temperature normalization
- Symptom-based user input integration for early health indicators

The system generates a unified risk score using a weighted model and provides real-time, location-based alerts to support decision-making for medical teams.

Unlike existing systems, the proposed solution focuses on:

1.Early detection instead of diagnosis
2.Multi-factor integration instead of isolated analysis
3.Privacy-preserving design without storing personal data
4.Suitability for large-scale environments such as Hajj

CHAPTER 3

PROBLEM STATEMENT AND OBJECTIVES

3.1 Operational Problem Statement

The Hajj environment suffers from many complex health and security challenges due to crowd density and high temperatures. Previous research has extensively explored crowd density using computer vision and deep learning [3, 4, 7], while other studies have focused on heat-related illnesses through clinical measurements and post-accident diagnosis [5, 6], symptom-based health systems have been studied mainly from a user experience perspective without incorporating environmental or crowd density-related risk factors [1, 2]. It was often developed independently, resulting in solutions that lack integration. Studies highlight a critical gap represented by the absence of integrated preventive systems that combine environmental risks associated with heat, congestion conditions and early health indicators in one framework suitable for high-congested environments such as Hajj.Current solutions focus on diagnostic rather than early warning and require complex clinical data or wearable devices and lack immediate spatial awareness and decision support mechanisms for medical response teams. The project aims to fill these gaps by developing an intelligent platform to monitor the health and safety of pilgrims in crowded and high-temperature pilgrimage environments. The proposed solution combines crowd density analysis using computer vision technology with a hypothetical analysis of heat risks and simple symptoms reported by pilgrims themselves, such as dizziness, fatigue or itching. The proposed system follows a hierarchical integration approach. First, a CNN-based crowd density model, drawing on recent studies [3, 7, 8], classifies each zone's density as low, medium, or high. Second, a heat risk simulation module computes a baseline environmental risk score using the formula: (Density × 0.6) + (T_{norm}× 0.4), where density is assigned a higher weight (60%) due to its dual impact on both mobility restriction and heat stress. Third, a rule-based engine incorporates self-reported symptoms by adding +1 to the score for a single report, and +2 for multiple reports (three or more) in the same geographical zone. The final score5determines the alert level: low (1.0-1.5), preventive warning (1.6-2.0), high risk (2.1-2.6), or critical (>2.6), which triggers an immediate notification to medical teams. The controlled environment is divided into pre-defined areas, allowing the system to link reported cases to specific locations (e.g. Zone A, Zone B). Spatial division helps identify high-risk areas and supports medical teams by providing location-based alerts and aggregated risk information. The system does not provide a medical diagnosis but focuses on early warning and decision support to help reduce health risks, prioritize medical response and enhance public safety during the Hajj season.

3.2 Functional Goals of the System

The project aims to:

1. Design and develop an intelligent software platform capable of monitoring health risks resulting from crowd congestion and high temperatures during the Hajj season.
2.Employ computer vision techniques to monitor crowd density and identify high- risk zones, targeting a classification accuracy of at least 85% when tested on benchmark datasets, consistent with findings reported in recent literature [3, 4].
3. Provide a hypothetical representation of heat stress risks based on crowd density levels and environmental conditions, using a formula that weights density more heavily (60%) due to its dual impact on both heat stress and mobility restriction, and 40% for normalized temperature.
4. Link crowd density and heat risks to self-reported symptoms through a rule-based engine that dynamically updates zone risk scores, triggering immediate notifications to medical teams when critical levels are detected.
5. Offer a simple user interface that accommodates diverse pilgrim demographics and supports medical teams in decision-making without requiring advanced medical diagnosis, with an end-to-end response time target (from image analysis to alert generation) of under 5 seconds in the prototype system.
6. Enable spatial risk monitoring by dividing Hajj ritual areas into predefined sectors, allowing medical teams to prioritize responses and allocate resources more efficiently.
7.Evaluate the integrated system's performance in a simulated environment by measuring the accuracy of risk level classification compared to validated reference scenarios, precision and recall rates in detecting high-risk areas, and user satisfaction (medical teams) regarding alert clarity through a simplified questionnaire.

3.3 Scope, Constraints, Assumptions

This project aims to develop an AI-based prototype to support early health and safety monitoring in crowded and high-temperature environments. The system focuses on analyzing heat- and crowd-related risk indicators using simple user-reported symptoms and general visual cues extracted from images, such as visible fatigue or skin redness. The project does not provide medical diagnosis, does not store personal data, and does not address chronic or complex medical conditions. It is limited to simple and potentially urgent cases related to heat and crowd exposure and will be implemented as a prototype.

The project is based on the following assumptions:

- The Hajj environment can be divided into predefined spatial zones for monitoring.
- Environmental temperature values can be obtained from historical weather data for Makkah or simulated within realistic ranges (30–45°C).
- Pilgrims are able to report simple symptoms through a web-based interface without technical training.
- The ShanghaiTech Part A dataset provides adequate representation of high-density crowd scenes similar to Hajj conditions.

CHAPTER 4

RESEARCH METHODOLOGY AND PLAN

4.1 System Development Methodology

This project follows an applied development methodology to build a web-based AI prototype for early health risk monitoring in crowded and high-temperature environments such as Hajj. The system integrates computer vision-based crowd density classification, heat-risk representation, and simple self-reported symptoms within predefined spatial zones. The crowd density model is evaluated using accuracy, precision, recall, and F1-score. The dataset is split into 70% training, 15% validation, and 15% testing to ensure proper generalization. Environmental temperature values are obtained from historical weather data for Makkah or simulated within realistic Hajj temperature ranges (30–45°C).

4.2 Tools, Platforms, and Technologies

The platform is implemented using Python, with OpenCV for image processing and CNN models developed using TensorFlow or PyTorch to classify density levels (low, medium, high). The output of the CNN-based model is mapped into numerical density levels (Low = 1, Medium = 2, High = 3), which are then used in the risk calculation formula. A lightweight web interface is developed using Flask or Streamlit. Heat-related risk is calculated using a weighted formula:

Baseline Risk Score = 0.6 (Density) + 0.4 (T_{norm})

Temperature values are normalized using:

T_{norm} = (T − T_min) / (T_max − T_min)

This model follows a weighted composite indicator approach as described in the OECD Handbook on Constructing Composite Indicators [10], while temperature contributes 0.4 as an environmental stress factor supported by heat-related health studies [9]. To incorporate health-related indicators, a rule-based symptom engine adjusts the risk score by adding +1 for a single symptom report and +2 for multiple reports (three or more) within the same geographical zone. Self-reported symptoms (e.g., dizziness, fatigue, itching) are collected through the system interface and used only to adjust the risk score without storing personal information. The final score is categorized into predefined alert levels (low, medium, high, or critical), which are displayed through an administrative monitoring interface.

4.3 Data Sources

The crowd density model is trained using the publicly available ShanghaiTech Part A dataset [11], which contains high-density crowd scenes representative of large-scale gatherings such as Hajj. If additional data is needed, the UCF-QNRF dataset will be used as a backup to improve generalization.

4.4 Ethical Considerations

The proposed system adheres to internationally recognized ethical standards in artificial intelligence, data governance, and public health. Given the sensitive context of mass gatherings such as Hajj, the design and implementation of this project prioritize privacy, fairness, transparency, and responsible use of technology.

1. Privacy and Data Protection

   The system does not collect, store, or process personally identifiable information. All analyses rely on publicly available datasets, such as the ShanghaiTech dataset, ensuring the protection of individual privacy and compliance with ethical data practices.

2. Responsible and Non-Diagnostic Use

   The proposed solution serves as an early risk detection and decision-support tool. It does not provide medical diagnoses or replace professional healthcare services. Instead, it generates preventive alerts to support medical teams and enhance pilgrim safety.

3. Transparency and Explainability

   The methodologies, algorithms, and data sources used in the system are clearly documented to ensure transparency and accountability. This promotes trust in AI-driven decision-making and aligns with global ethical AI principles.

4. Fairness and Bias Mitigation

   The system utilizes standardized benchmark datasets and validated evaluation metrics to minimize bias and ensure equitable performance across diverse populations. Continuous evaluation and testing are conducted to maintain reliability and fairness.

5. Data Security and Confidentiality

   Any environmental or symptom-related data are handled securely and used solely for risk assessment. No personal or medical data are stored or shared, ensuring confidentiality and compliance with ethical standards.

6. Compliance with International and National Ethical Frameworks

   The project aligns with globally recognized AI ethics principles, including those established by UNESCO, the OECD, and the European Commission, as well as national guidelines issued by the Saudi Data and Artificial Intelligence Authority (SDAIA).

The proposed system complies with international ethical standards in artificial intelligence and data governance, aligning with guidelines from UNESCO, OECD, the European Commission, SDAIA, and the World Health Organization [26]– [30].

4.5 Risk Assessment

To ensure the success of the project, several potential risks have been identified with appropriate mitigation strategies. Technically, the main risk is that the 85% target accuracy for the crowd model may not be achieved. To address this, pre-trained models will be used along with data augmentation techniques and hyperparameter tuning. Another risk is that the dataset may not adequately represent actual congestion conditions in the Hajj environment, in which case additional datasets such as UCF-QNRF will be used. Regarding environmental data, the lack of accurate temperature measurements for specific pilgrimage areas is a potential challenge. This will be addressed by simulating temperature values within realistic ranges (30–45°C) based on historical weather data for Mecca. Integration challenges between different system units may also arise; therefore, a modular design approach will be adopted with early integration testing. On the other hand, difficulty in achieving the target response time of under 5 seconds may be faced due to model complexity. This will be resolved by optimizing model efficiency and reducing input image size while maintaining accuracy. Finally, limited access to high-performance computational resources will be mitigated by using cloud-based platforms such as Google Colab and Kaggle.

4.6 Milestone-Based Project Schedule

Table 3: Project Timeline

Week	Phase	Activities	Deliverable
1	Project Initiation	Define project scope, objectives, and team roles	-
2	Background Study	Study Hajj health risks, crowd density estimation, and heat stress concepts	-
3	Literature Search	Collect research papers related to CNN crowd analysis, heat risk, and symptom-based systems	-
4	Literature Review	Review and summarize literature and identify research gap	Project Proposal
5-6	Project Planning	Prepare timeline, milestones, task allocation, and risk assessment	Project Management Plan
7	Report Writing	Write GP-I Report Chapters 1-2 (Introduction, Background)	-
8	Requirements	Document functional & non-functional requirements, risk formula	System Requirements
9	Report Writing	Write GP-I Report Chapters 3-4 (Problem Statement, Methodology)	-
10	Report Writing	Write GP-I Report Chapter 5 (Requirements, Use-Case)	-
11	Complete Chapter 5	System architecture, algorithms, design decisions, testing strategy (5.4-5.7)	-
12	Report Review	Review and refine all chapters	-
13	Report Finalization	Final formatting and proofreading	-
14	Submission	Prepare final GP-I Progress Report	GP-I Progress Report
15	Presentation Preparation	Prepare slides and rehearse oral presentation	-
16	Final Presentation	Present proposal, planning, and progress	GP-I Oral Presentation

This project is divided into a set of clearly defined milestones to ensure systematic development and timely completion of the proposed system.

Milestone 1: Project Planning and Requirement Definition

This milestone focuses on defining the project scope, understanding the problem domain, and identifying functional and non-functional requirements. The project objectives are finalized to ensure alignment with the intended outcomes. (Weeks 1-2)

Milestone 2: Literature Review and Gap Identification

Relevant studies related to crowd density estimation, heat-related health risks, and symptom-based health systems are reviewed. Existing approaches are analyzed to identify limitations and research gaps addressed by this project. (Weeks 3-4)

Milestone 3: Project Proposal Submission

The project proposal is completed, including background, scope, description, objectives, timeline, expected outcomes, methodology, and references. (Week 4)

Milestone 4: Project Management Plan Submission

The project management plan is prepared, including timeline, milestones, task allocation, and risk assessment. (Weeks 5-6)

Milestone 5: Report Writing (Chapters 1-2)

The GP-I report chapters are written, including Introduction and Motivation (Chapter 1) and Background and Related Work (Chapter 2). (Week 7)

Milestone 6: System Requirements and Design Submission

Functional and non-functional requirements are documented. (Week 8)

Milestone 7: Report Writing (Chapters 3-4)

The GP-I report chapters are written, including Problem Statement and Objectives (Chapter 3) and Methodology and Project Plan (Chapter 4). (Week 9)

Milestone 8: Report Writing (Chapter 5 - Part 1)

The GP-I report Chapter 5 is written, including Functional Requirements, Non-Functional Requirements, and Use-Case Analysis (Sections 5.1-5.3). (Week 10)

Milestone 9: Chapter 5 Completion

The remaining sections of Chapter 5 are completed, including System Architecture (5.4), High-Level Algorithms and Models (5.5), Design Decisions and Justification (5.6), and Planned Testing and Evaluation Strategy (5.7). (Week 11)

Milestone 10: Report Review and Finalization

All chapters are reviewed, refined, and formatted. Final proofreading is completed. (Weeks 12-13)

Milestone 11: GP-I Progress Report Submission

The final GP-I Progress Report is prepared and submitted. (Week 14)

Milestone 12: GP-I Oral Presentation

Presentation slides are prepared, and the oral presentation is delivered to the supervisor and examiner. (Weeks 15-16)

CHAPTER 5

SYSTEM ANALYSIS AND SYSTEM DESIGN

5.1 Functional Requirements

1. Crowd Density Classification

The system shall analyze crowd images and classify density levels as Low, Medium, or High using a CNN-based model to support medical teams and authorities in identifying overcrowded areas quickly and taking early preventive actions to reduce risks.

2. Zone-Based Monitoring

Target Users: Authorities

The system shall organize the monitored area into predefined zones (such as Zone A and Zone B), enabling authorities to track risk levels for each location and prioritize high-risk zones for timely intervention.

3. Temperature Retrieval and Normalization

The system shall retrieve environmental temperature values from historical weather data or simulated inputs and normalize them before including them in the risk calculation process, allowing medical teams and authorities to obtain accurate heat risk assessments and make informed decisions.

4. Baseline Risk Score Calculation

The system shall calculate a baseline risk score by combining crowd density levels with normalized temperature values using a predefined weighted formula, providing medical teams and authorities with a clear and measurable indicator of risk in each zone.

5. Symptom Reporting Interface

Target Users: Pilgrims

The system shall allow pilgrims to report simple symptoms, such as dizziness, fatigue, or itching, through a user-friendly web interface, supporting early detection of potential health risks and increasing awareness.

6. Rule-Based Risk Adjustment

The system shall dynamically adjust the risk score of each zone based on the number of symptoms reported by users in the same area, enabling medical teams and authorities to obtain accurate and real-time assessments of risk conditions.

7. Alert Generation and Dashboard Visualization

Target Users: Medical Teams, Authorities, and Pilgrims

The system shall generate alert levels (Low, Preventive Warning, High Risk, or Critical) and display them on an administrative dashboard for medical teams and authorities. Additionally, the system shall provide preventive notifications to pilgrims in high-risk zones to encourage safer behavior.

5.2 Non-Functional Requirements

1. Performance
The system must be fast in processing and issuing alerts within 5 seconds from the time of image analysis to sending the alert to ensure rapid response in critical cases

2. Reliability
The system must achieve 99% uptime during continuous 8-hour testing periods, with no more than one unexpected failure per 1,000 user requests.

3. Scalability
The system must support at least 500 concurrent users and process up to 100 symptom reports per minute while maintaining the 5-second response time.

4. Ease of use
The system interface must be usable and clear for pilgrims without technical training, and medical teams, requiring no more than two clicks to report a symptom.

5. Security and Privacy
The system must store no personally identifiable information (PII). All symptom reports must be anonymized and aggregated at the zone level only.

6. Maintainability
The system will be designed in a divided way (Modular) so that we can update the model and fix errors or develop a specific part that we can easily do without ruining the rest of the system

7. Availability
The system is working and available at least 99% all the time to ensure continuous risk monitoring in a high-temperat environmen

5.3 Use-Case Analysis

Use Case Description:

The use case diagram illustrates the interaction between the user's pilgrims, medical teams, and authorities and the proposed AI-based early risk detection system. Pilgrims can report symptoms and receive preventive notifications, while medical teams and authorities monitor system outputs through the dashboard and receive alerts for high-risk zones. The system processes crowd density, temperature data, and user-reported symptoms to calculate risk levels and generate alerts, supporting timely decision-making in crowded environments such as Hajj.

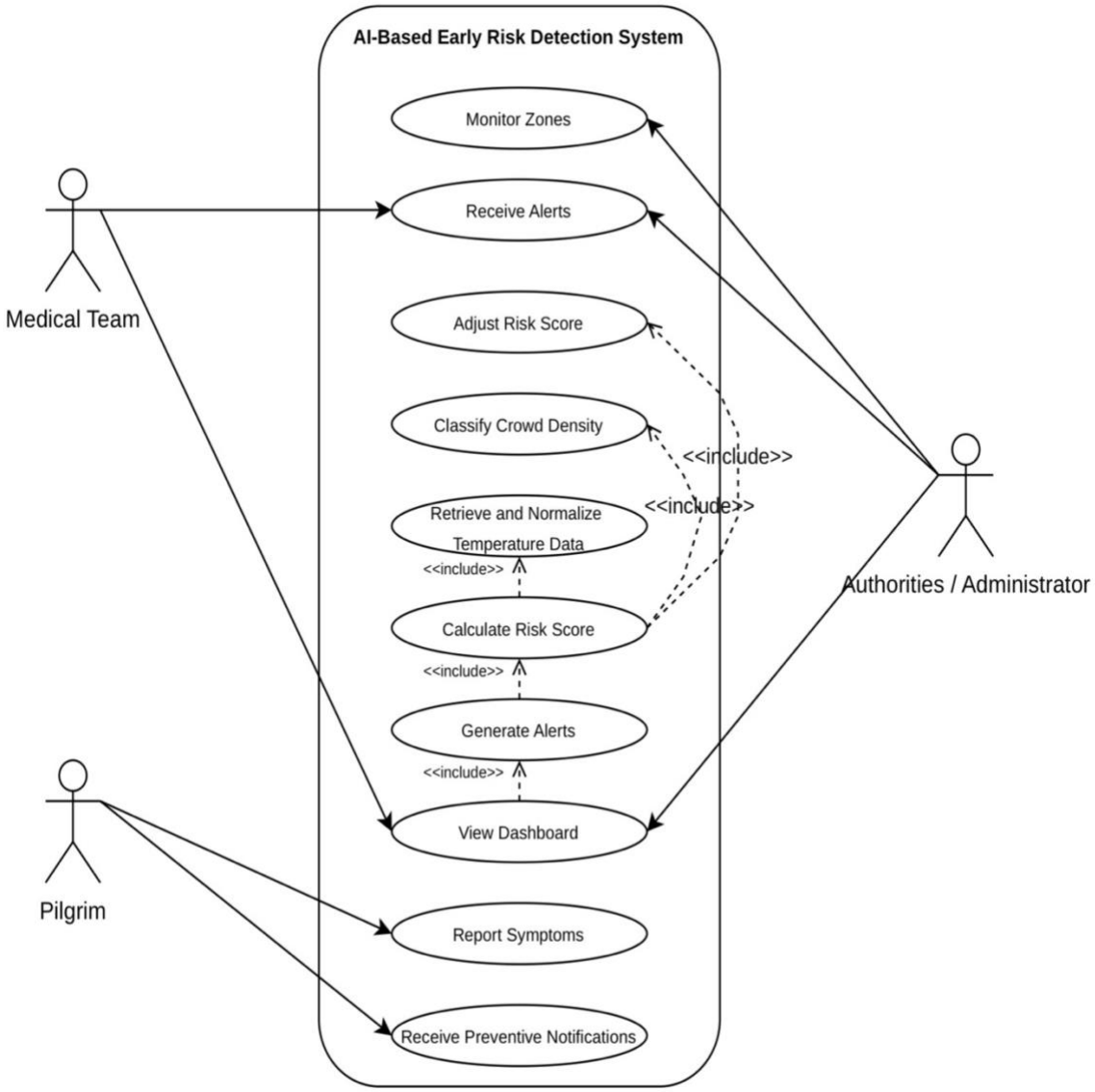

Figure 1 Use Case Diagram

5.4 User Scenario

Scenario 1: Pilgrim Reporting Symptoms

A pilgrim located in a crowded area (e.g., Zone A) begins to feel slight symptoms such as dizziness and fatigue because of the high temperature and crowd density. Using the system's web interface, the pilgrim quickly reports these symptoms by choosing them from a predefined list. The system records the input anonymously and associates it with the corresponding zone. Based on the accumulation of symptom reports in the same area, the system adjusts the risk score accordingly. If the risk level increases, the pilgrim receives a preventive notification advising them to rest, hydrate, or move to a less crowded area.

Scenario 2: High-Risk Zone Monitoring by Authorities

In this scenario, authorities monitor different zones through the administrative dashboard, where they observe real-time crowd conditions. Incoming images are continuously analyzed using the CNN model to classify crowd density levels, while temperature data is collected and processed for each zone. Based on these inputs, the system calculates the baseline risk level for every zone. When a specific zone reaches a high-risk or critical level, the system automatically generates an alert to support timely intervention.

Scenario 3: Medical Team Responding to Alert

The medical team monitors the system dashboard during peak hours. The system detects high crowd density combined with elevated temperature and multiple symptom submissions in Zone B. As a result, the risk score exceeds the critical threshold, triggering an alert. Consequently, the medical team is immediately notified and can quickly identify the affected zone. Based on this information, they can take appropriate measures to handle potential cases related to extreme heat.

5.5 Entity-Relationship Digram (ER Digram)

The database consists of four main entities: pilgrim, zones, symptoms, and alerts. Pilgrim and zones have a many-to-one relationship (m:1), where multiple pilgrims can be located in one zone. symptoms have many-to-one relationships with both pilgrim and zones, as multiple symptoms can be reported by one pilgrim in one zone. alerts have a many-to-one relationship with zones, where multiple alerts can be generated for a single zone.

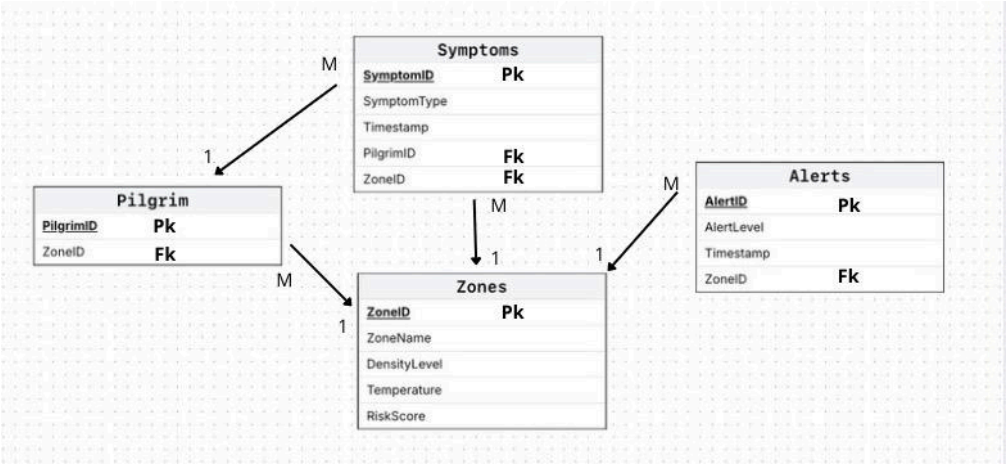

Figure 2: ER Diagram

5.6 Database Schema

TABLE NAME	COLUMN NAME	DATA TYPE	KEY	DESCRIPTION
Pilgrim	PilgrimID	INT	PRIMARY KEY	Unique pilgrim identifier
	ZoneID	INT	FOREIGN KEY	References zone table
ZONES	ZoneID	INT	PRIMARY KEY	Unique zone identifier
	ZoneName	VARCHAR (10)	-	Zone name A, B, C
	DensityLevel	INT	-	1=Low, 2=Medium, 3=High
	Temperature	FLOAT	-	Temperature
	RiskScore	FLOAT	-	Calculated risk score
SYMPTOMS	SymptomID	INT	PRIMARY KEY	Unique symptom report identifier
	SymptomType	VARCHAR (20)	-	dizziness, fatigue, itching
	Timestamp	DATETIME	-	Time of symptom report
	PilgrimID	INT	FOREIGN KEY	References pilgrim table
	ZoneID	INT	FOREIGN KEY	References zones table
ALERTS	AlertID	INT	PRIMARY KEY	Unique alert identifier
	AlertLevel	VARCHAR (20)	-	Low, Warning, High, Critical
	Timestamp	DATETIME	-	Time Alert Was generated
	Status	VARCHAR (20)	-	Sent, Acknowledged
	ZoneID	INT	FOREIGN KEY	References Zones table

Table 4: Database Schema

5.7 Class Digram

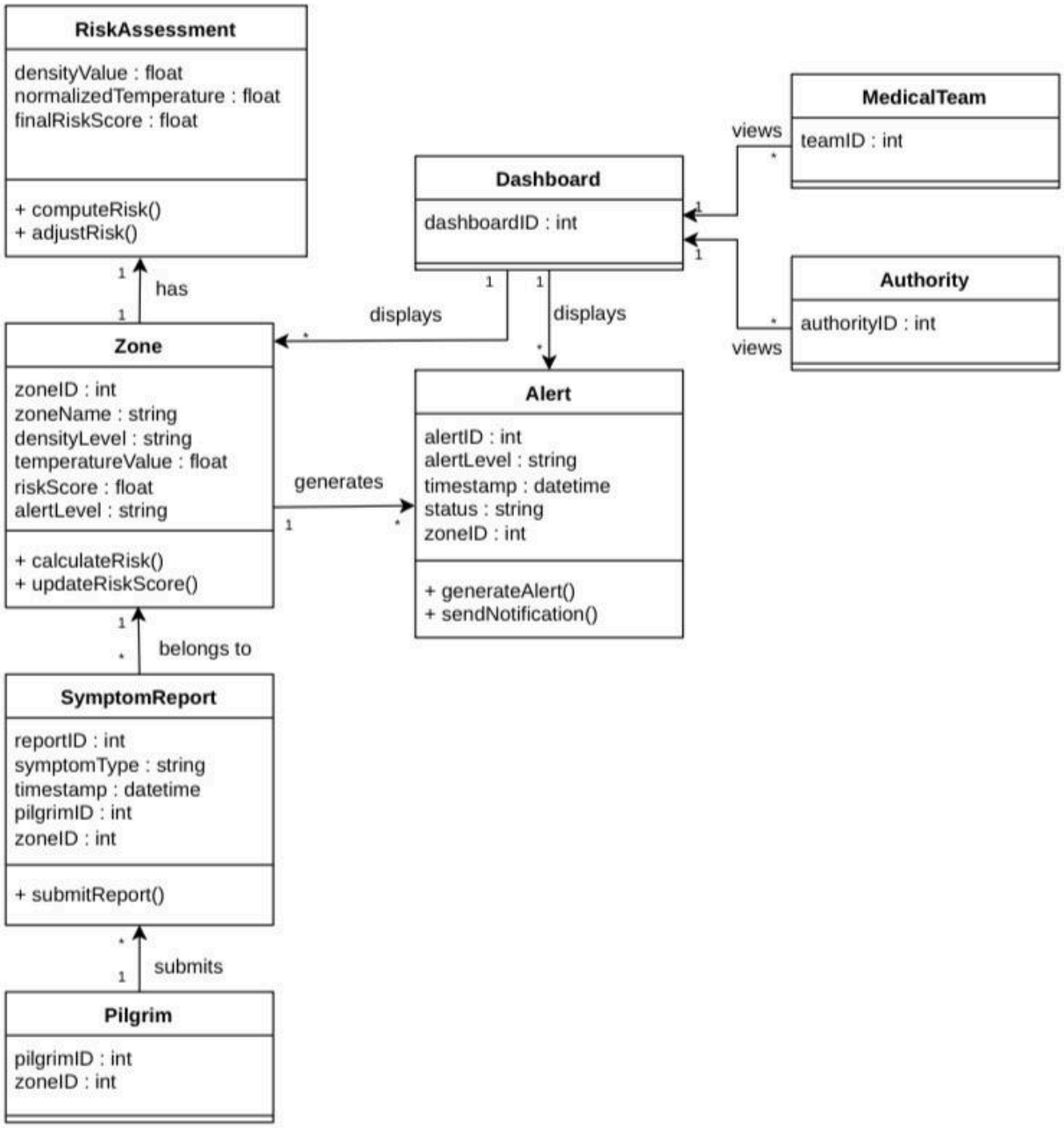

Figure 3: Class Diagram

5.8 Data Flow Digram

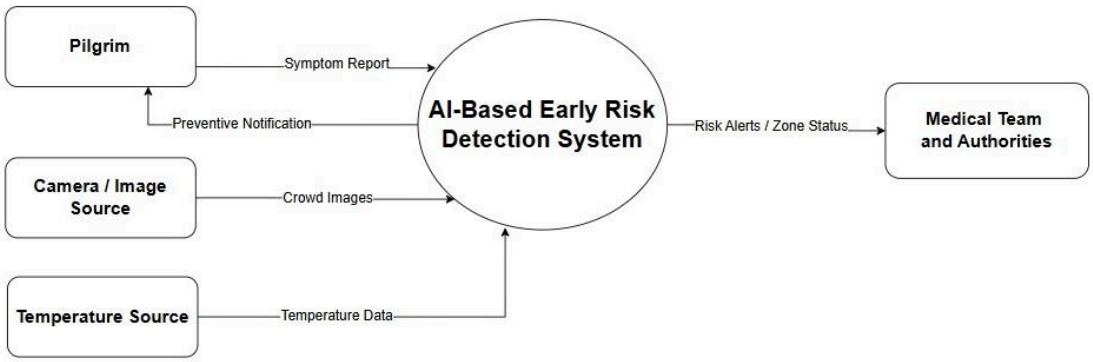

Figure 4: Data Flow Diagram Level 0

This diagram represents the Level 0 Data Flow Diagram, also known as the Context Diagram, for the AI-Based Early Risk Detection System. It shows the overall interaction between the system and the external entities without presenting the internal processes. The Pilgrim sends a Symptom Report to the system and receives a Preventive Notification in return. The Camera or Image Source provides Crowd Images, while the Temperature Source sends Temperature Data. All these inputs are processed by the system internally. After analyzing the received data, the system sends Risk Alerts and Zone Status information to the Medical Team and Authorities. Therefore, this diagram provides a high-level view of how data enters and leaves the system and how the system communicates with external actors.

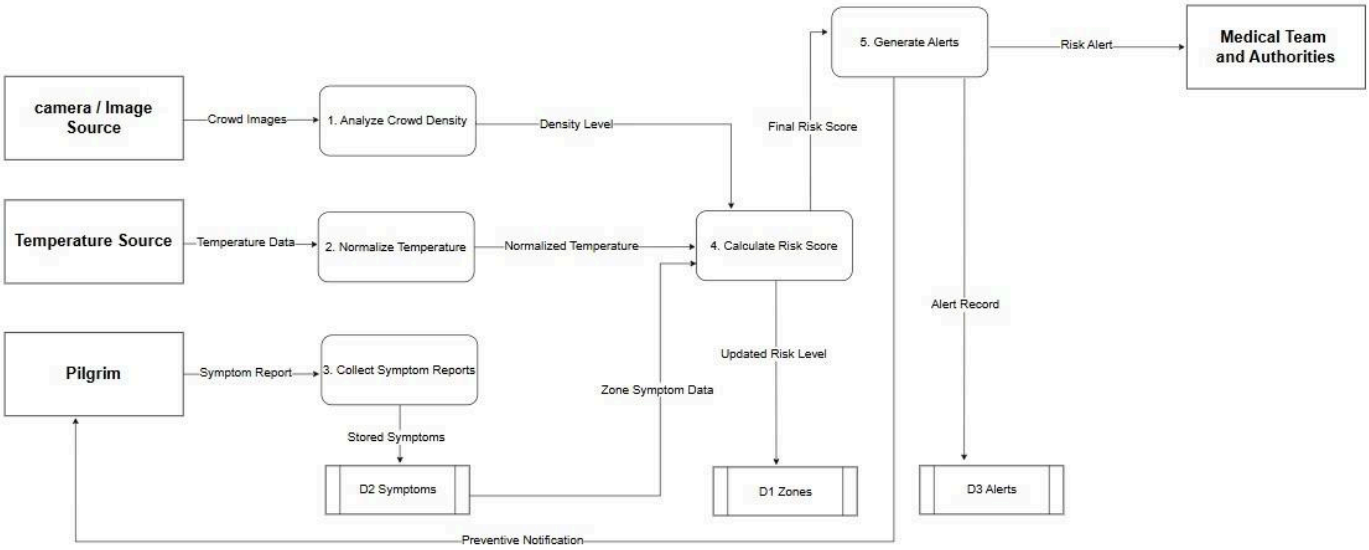

Figure 5: Data Flow Diagram Level 1

This diagram represents the Level 1 Data Flow Diagram (DFD) for the AI-Based Early Risk Detection System. It illustrates how the system operates internally by dividing it into a set of interconnected processes that work together to transform input data into meaningful outputs.

The system begins by receiving inputs from external entities. The Camera or Image Source provides crowd images to the Analyze Crowd Density process, which determines the crowd density level. At the same time, the Temperature Source sends temperature data to the Normalize Temperature process, where it is converted into a normalized value suitable for further processing. In addition, the Pilgrim sends a Symptom Report to the Collect Symptom Reports process, where this information is stored in the Symptoms Data Base (D2 Symptoms).

The outputs from these processes, including Density Level, Normalized Temperature, and Zone Symptom Data, are then forwarded to the core process, Calculate Risk Score. This process integrates all inputs to compute the final risk level for each zone.

Once the risk level is calculated, it is stored in the Zones Data Base (D1 Zones). The Final Risk Score is then passed to the Generate Alerts process, which is responsible for producing alerts based on the calculated risk level.

The generated alerts are stored in the Alerts Data Base (D3 Alerts) as Alert Records. At the same time, Risk Alerts are sent to the Medical Team and Authorities to support decision-making and enable timely intervention. Additionally, Preventive Notifications are sent to the Pilgrim to help reduce exposure to potential risks.

Overall, this diagram demonstrates how the system integrates multiple data sources, including visual data, environmental data, and user-reported symptoms, to generate effective alerts and support early risk detection in crowded environments.

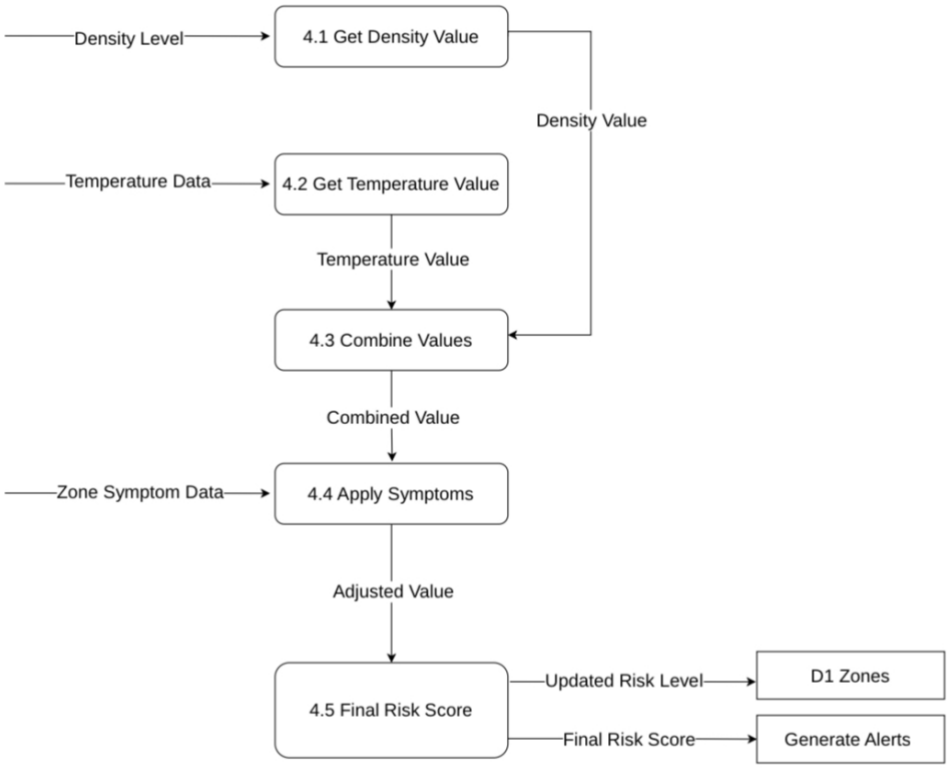

Figure 6: Data Flow Diagram Level 2

This diagram represents the Level 2 Data Flow Diagram (DFD) of the Calculate Risk Score process in the AI-Based Early Risk Detection System. It illustrates how the risk calculation process is decomposed into smaller steps, where data is processed gradually until the final result is produced.

The process begins by receiving inputs from Level 1. The Density Level is sent to 4.1 Get Density Value, where it is converted into a usable density value. At the same time, Temperature Data is sent to 4.2 Get Temperature Value, where it is transformed into a temperature value suitable for processing.

The outputs of these two processes, Density Value and Temperature Value, are then passed to 4.3 Combine Values, where they are integrated to produce a combined value that reflects the impact of both crowd density and temperature on the risk level.

Next, the combined value is sent to 4.4 Apply Symptoms, which also receives Zone Symptom Data (from Level 1). In this step, the system adjusts the risk level based on reported symptoms, making the evaluation more accurate and context aware.

Finally, the adjusted value is passed to 4.5 Final Risk Score, which represents the last stage of the process, where the system generates the final risk score for each zone.

The result is then used in two ways: it is stored in D1 Zones as an Updated Risk Level, and it is also sent to the Generate Alerts process as the Final Risk Score, enabling the system to produce alerts and support timely decision-making.

Overall, this diagram demonstrates how the system integrates crowd density, temperature, and symptom data to generate an accurate and dynamic risk assessment.

5.9 System Architecture

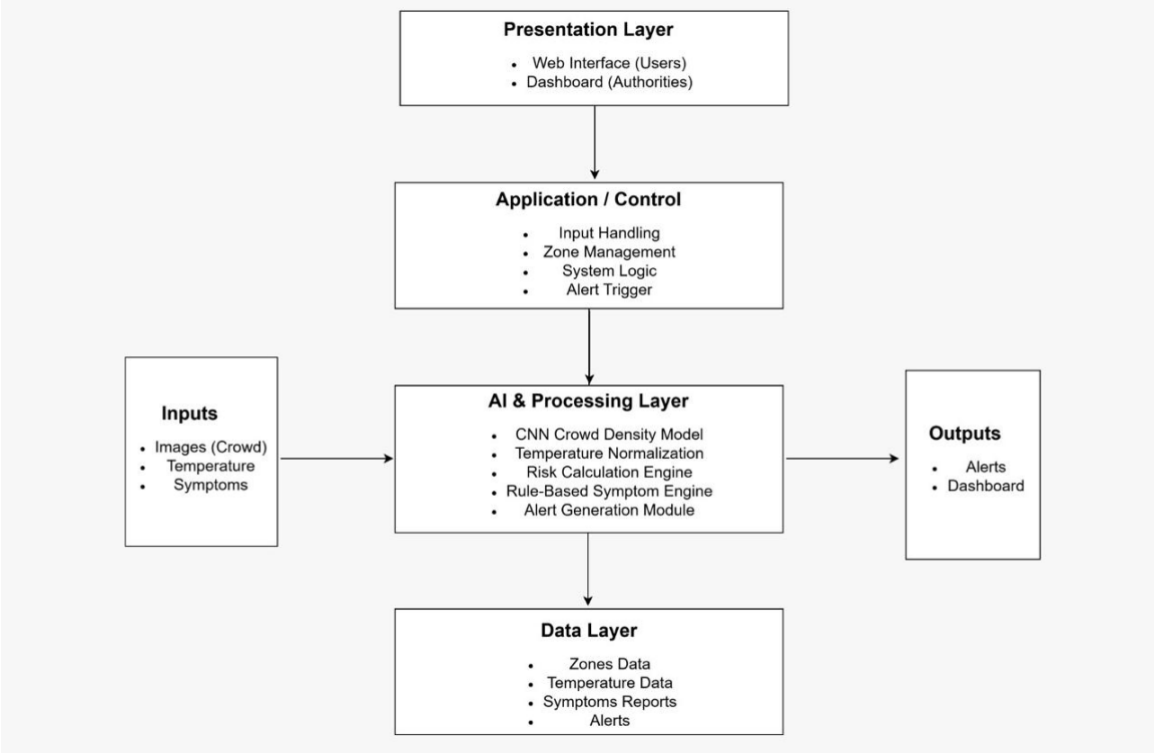

Figure 7: System Architecture

This figure illustrates the system architecture of the AI-Based Early Risk Detection System, which follows a layered design to ensure clear organization and efficient data processing. The architecture is divided into four main layers, each responsible for a specific function within the system.

At the top, the Presentation Layer represents the user interface, including the web interface used by pilgrims to report symptoms and receive notifications, as well as the dashboard used by authorities to monitor risk levels and system outputs. This layer is responsible for displaying information and enabling user interaction.

Below it, the Application / Control Layer manages the system logic and controls the overall workflow. It handles user inputs, manages zones, processes system rules, and determines when alerts should be triggered based on the analyzed data.

The core of the system lies in the AI & Processing Layer, where data is analyzed and decisions are made. In this layer, crowd images are processed using a CNN-based model to estimate density, temperature data is normalized, and a risk calculation engine combines these inputs. Additionally, a rule-based symptom engine adjusts the risk level based on reported symptoms, and an alert generation module produces the appropriate alerts.

At the bottom, the Data Layer is responsible for storing all system data, including zones information, temperature data, symptom reports, and generated alerts. This ensures data availability for processing and future analysis.

The system receives inputs such as crowd images, temperature data, and user-reported symptoms. These inputs are processed through the different layers to produce outputs in the form of alerts and dashboard visualizations, enabling authorities to take timely and informed actions.

Overall, this layered architecture provides a structured, scalable, and maintainable design, allowing the system to efficiently integrate multiple data sources and deliver accurate real-time risk assessments.

5.10 High-Level Algorithms and Models

The system is based on two main models of risk analysis and classification:

First: Crowd Density Classification Model

A Convolutional Neural Network (CNN) is used to analyze images taken from different Hajj areas. It performs the following:

-Receives an image of a specific area (e.g. region A or B)

-Analyze the features of the image to detect the number of people and their distribution

- Classifies the region into one of three categories: Low Density, Medium Density, or High Density

The model will be trained using the ShanghaiTech Part A dataset, a collection containing high-density crowd scenes similar to the Hajj environment, targeting a classification accuracy of at least 85%.

Second: Risk Calculation and Adjustment Model

After obtaining the density level, the system calculates the risk score using a weighted formula:

Baseline Risk Score = (Density × 0.6) + (Normalized Temperature × 0.4)

Where:

-Density: numerical value (1 = low, 2 = medium, 3 = high)

-Normal temperature: a value between 0 and 1 calculated by :$\mathbf{T_{norm} = (T - T_min) / (T_max - T_min)}$

Where T_min = 30°C and T_max = 45°C (typical Hajj temperature range).

Density has been given a higher weight (60%) because it has a dual effect: it increases heat stress and makes movement and ventilation more difficult.

Symptom-Based Risk Adjustment (Rule-Based Engine)

After calculating the baseline risk score, the rule-based engine adjusts the score based on reports received from pilgrims in the same area:

+1 for a single symptom report (dizziness, fatigue, itching)

+2 for three or more reports in the same region

Final Risk Score and Alert Levels

After adjustment, the final score determines the alert level:

1.0 - 1.5 Low (No action)

1.6 - 2.0 Preventive Warning (Send notification to pilgrims)

2.1 - 2.6 High Risk (Send strong warning)

More than 2.6 Critical (Immediate notification to medical teams)

5.11 Design Decisions and Justification

This section presents the key design decisions made in developing the proposed system, along with clear justifications based on the project requirements, identified research gaps, and practical constraints.

1. Use of CNN for Crowd Density Classification

A Convolutional Neural Network (CNN) is used to classify crowd density levels (Low, Medium, High).

Justification:

CNNs are widely recognized for their strong performance in image analysis and crowd density estimation, especially in complex and dense environments. Compared to traditional image processing techniques, CNNs can automatically extract spatial features and handle occlusion more effectively. This decision directly addresses the need for accurate real-time crowd analysis, which is essential for risk assessment in crowded environments like Hajj. Additionally, CNN-based models were shown in prior studies to achieve high accuracy in dense scenes, making them suitable for this application.

2. Multi-Factor Risk Integration

The system integrates three main factors:

- Crowd density
- Environmental temperature
- User-reported symptoms

Justification:

Most existing systems focus on a single aspect (either crowd, health, or environment), leading to incomplete risk assessment. This limitation was clearly identified as a research gap in the literature review. By combining these factors into a unified model, the system provides a more comprehensive and realistic understanding of risk, improving early detection and decision-making.

3. Weighted Risk Formula

The baseline risk score is calculated using a weighted formula that assigns 60% to crowd density and 40% to temperature.

Justification:

The weighting approach follows composite indicator methodologies, where multiple factors are assigned different importance levels based on their impact [10]. Temperature is supported in the literature as a key environmental risk factor in heat-related conditions, especially in mass gatherings such as Hajj [18]. Crowd density was assigned a higher weight because it has a dual effect: it increases heat stress and restricts movement and ventilation, making it a more critical factor in real-world risk scenarios. Therefore, the selected weights (0.6 and 0.4) were defined as a design decision to better reflect the combined impact of environmental and physical factors on risk.

4. Use of Rule-Based Engine for Symptoms

A rule-based system is used to adjust risk scores based on symptom reports.

Justification:

Instead of using complex medical AI models, a rule-based approach was chosen because:

- The system is designed for early warning, not medical diagnosis
- It reduces computational complexity
- It improves interpretability and transparency

This aligns with the project goal of providing a lightweight and practical prototype while still incorporating health-related indicators effectively.

5. Zone-Based System Design

The monitored environment is divided into predefined zones (e.g., Zone A, Zone B).

Justification:

Crowded environments such as Hajj are spatially complex and require more organized monitoring. A zone-based approach enables localized risk detection by linking data to specific areas, which helps medical teams respond more quickly and allocate resources more effectively. This design improves system scalability and makes the output more actionable compared to global analysis.

6. Web-Based Lightweight Architecture

The system is implemented as a lightweight web-based application using frameworks such as Flask or Streamlit.

Justification:

A web-based solution ensures easy accessibility for different users, including pilgrims, authorities, and medical teams, without requiring installation. It also allows faster deployment and testing of the system and adopting a lightweight architecture supports the system requirement of achieving a fast response time of under five seconds, which is essential for real-time alert generation in critical situations.

7. Privacy-Preserving Design

The system does not store any personal or identifiable user data.

Justification:

Privacy is a critical concern in large-scale public systems, especially in sensitive environments such as Hajj. By avoiding the storage of personal data, the system minimizes ethical risks, ensures compliance with data protection standards, and increases user trust. This design choice aligns with internationally recognized AI ethics and data governance principles, which emphasize privacy, transparency, and responsible data usage [26]–[30].

8. Use of Public Dataset (ShanghaiTech Part A)

The ShanghaiTech dataset is used for training the crowd density model.

Justification:

Using a publicly available dataset saves time and effort compared to collecting custom data. The dataset contains high-density crowd scenes similar to real-world environments such as Hajj, making it suitable for training and evaluation.

9. Modular System Design

The system is designed using a modular structure, separating components such as vision processing, risk calculation, and user interface.

Justification:

This design makes the system easier to manage because each part works independently and can be updated without affecting the others. It also helps in finding and fixing errors more quickly. In addition, it allows the system to be expanded in the future by adding new features easily. This is useful in AI systems, where models may need updates or improvements over time.

5.12 Planned Testing and Evaluation Strategy

To ensure that the proposed AI-based early risk detection system operates accurately, efficiently, and reliably, a comprehensive testing and evaluation strategy will be implemented. This strategy aims to validate both the functional and non-functional aspects of the system, in addition to assessing the performance of the underlying machine learning model.

Functional testing will be conducted to verify that all system components operate according to the specified requirements. The crowd density classification model will be evaluated using unseen images to ensure its ability to correctly classify density levels into Low, Medium, and High categories. In addition, the zone-based monitoring feature will be tested to confirm the correct assignment and tracking of predefined zones and their associated risk levels. The system will also be validated to ensure accurate retrieval and normalization of temperature values based on the defined formula. Furthermore, the baseline risk score will be validated by comparing system-generated outputs with manually computed expected values to ensure correctness and consistency. The symptom reporting interface will be tested to confirm that users can easily submit symptoms and that the data is recorded accurately. Finally, alert generation will be verified to ensure that the correct alert levels are triggered based on the calculated risk scores.

In terms of model evaluation, the performance of the crowd density classification model will be assessed using standard evaluation metrics. The model is expected to achieve a minimum accuracy of 85%, while precision and recall will be measured to ensure balanced performance across all density classes. A confusion matrix will also be used to analyze misclassification patterns and identify potential weaknesses in the model. To ensure proper generalization and avoid overfitting, the dataset will be divided into training, validation, and testing subsets.

Performance testing will be carried out to ensure that the system meets the required response time constraints. Specifically, the system must be capable of generating alerts within 5 seconds from the time an image is processed. This will be evaluated under both normal and peak operating conditions. In addition, scalability testing will be performed to assess the system's ability to handle increased workloads, including supporting at least 500 concurrent users and processing up to 100 symptom reports per minute without significant degradation in performance.

Reliability testing will focus on ensuring system stability during continuous operation. The system will be tested over an extended period of at least 8 hours to verify that it maintains consistent performance, with a failure rate not exceeding one failure per 1,000 requests. Usability testing will also be conducted with representative users, such as pilgrims and medical staff, to evaluate the ease of use of the system interface. This includes ensuring that symptom reporting can be completed within two clicks and that the interface is clear and user-friendly.

Moreover, security and privacy testing will be conducted to ensure that the system complies with data protection requirements. The system will be verified to ensure that no personally identifiable information (PII) is stored, and that all symptom data is properly anonymized and aggregated at the zone level. Secure data handling practices will also be evaluated.

Finally, end-to-end system testing will be performed by simulating real-world scenarios in which crowd images, temperature data, and symptom reports are processed together. The system will be evaluated based on its ability to accurately calculate risk levels and generate appropriate alerts and dashboard visualizations. This ensures that all system components function cohesively to deliver reliable and timely risk assessments in crowded environments such as Hajj.

REFERENCES

[1] K. Arellano Carmona, D. Chittamuru, R. L. Kravitz, S. Ramondt, and A. S. Ramírez, "Health Information Seeking From an Intelligent Web-Based Symptom Checker: Cross-Sectional Questionnaire Study," J. Med. Internet Res., vol.24, no. 8, p. e36322, Aug. 2022, doi: 10.2196/36322.

[2] A. W. Liu, A. Y. Odisho, W. Brown III, R. Gonzales, A. B. Neinstein, and T. J. Judson, "Patient Experience and Feedback After Using an Electronic Health Record-Integrated COVID-19 Symptom Checker: Survey Study," JMIR Hum. Factors, vol. 9, no. 3, p. e40064, Sep. 2022, doi: 10.2196/40064.

[3] W. Mansouri, M. A. Alohali, H. Alqahtani, N. Alruwais, M. Alshammeri, and A. Mahmud, "Deep convolutional neural network-based enhanced crowd density monitoring for intelligent urban planning on smart cities," Sci. Rep., vol. 15, art. no. 5759, 2025, doi: 10.1038/s41598-025-90430-4.

[4] D. Mabrouk, M. A. Abdel-Fattah, and A. Taha, "Robust crowd anomaly detection via hybrid ensemble learning for real-world surveillance," Sci. Rep., vol. 15, art. no. 41832, 2025, doi: 10.1038/s41598-025-27408-9.

[5] W. Alkassas, A. M. Rajab, S. T. Alrashood, M. A. Khan, M. Dibas, and M. Zaman, "Heat-related illnesses in a mass gathering event and the necessity for newer diagnostic criteria: a field study," Environ. Sci. Pollut. Res., vol. 28, pp. 16682–16689, 2021, doi: 10.1007/s11356-020-12154-4.

[6] S. Yezli, S. Ehaideb, Y. Yassin, B. Alotaibi, and A. Bouchama, "Escalating climate-related health risks for Hajj pilgrims to Mecca," J. Travel Med., vol. 31, no. 4, taae042, Mar. 2024, doi: 10.1093/jtm/taae042.

[7] A. A. Alharbi, "DeepCAMS: A Deep Learning Approach for Real-Time Crowd Monitoring and Suspicious Behavior Detection Using Spatial-Temporal Analysis," Eng. Technol. Appl. Sci. Res., vol. 15, no. 4, pp. 26113– 26119, 2025, doi: 10.48084/etasr.10954.

[8] P. Kannan, V. Jadhav, Y. Patil, K. Hyalij, and P. Bacchav, "Real-time crowd monitoring and management," International Research Journal of Advanced Engineering and Humanities (IRJAEH), 2025. doi: 10.47392/IRJAEH.2025.0135.

[9] World Health Organization, Health Factors Involved in Working under Conditions of Heat Stress: Report of a WHO Scientific Group, Technical Report Series No. 412, Geneva, Switzerland: WHO, 1969. [Online]. Available: https://iris.who.int/handle/10665/40716

[10] OECD/European Union/Joint Research Centre, Handbook on Constructing Composite Indicators: Methodology and User Guide, Paris: OECD Publishing, 2008. [Online]. Available: https://www.oecd.org/en/publications/handbook-on-constructing-composite-indicators-methodology- and-user-guide_9789264043466-en.html

[11] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, "Single-Image Crowd Counting via Multi-Column Convolutional Neural Network," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 589–597

[12] A. B. Chan, Z. Liang, and N. Vasconcelos, "A Survey of Crowd Counting and Density Estimation Methods," IEEE Transactions on Pattern Analysis and Machine Intelligence, 2021. doi: 10.1109/TPAMI.2021.3053241.

[13] G. Wang X. Zhang, and H. Ma, "Deep Learning for Crowd Analysis: A Survey," ACM Computing Surveys, vol. 54, no. 4, 2021. doi: 10.1145/3460359.

[14] S. Shao, Z. Zhao, B. Li, and H. Wang, "TransCrowd: Weakly-Supervised Crowd Counting with Transformers," in Proc. ACM Multimedia (ACM MM), 2022. doi: 10.1145/3503161.3547860.

[15] Y. Zhang, D. Chan, and J. Xie, "Scale-Adaptive Convolutional Neural Network for Crowd Counting," IEEE Transactions on Circuits and Systems for Video Technology, 2020. doi: 10.1109/TCSVT.2020.3047811.

[16] M. Hossain, M. Islam, and R. Karim, "A Comprehensive Survey on Smart Healthcare Monitoring Systems Using IoT," Sensors, vol. 23, no. 5, 2023. doi: 10.3390/s23052551.

[17] S. S. Alghamdi and M. A. Khan, "Smart Healthcare Monitoring Systems: A Review of AI and IoT Integration," IEEE Access, 2022. doi: 10.1109/ACCESS.2022.3141390.

[18] World Health Organization, "Public Health for Mass Gatherings: Key Considerations," WHO, 2022. Available: https://www.who.int

[19] A. Bouchama and J. P. Knochel, "Heat Stroke," New England Journal of Medicine, 2020. doi: 10.1056/NEJMra1903298.

[20] S. Yezli and S. Khan, "Climate Change and Health Risks during Hajj," Journal of Travel Medicine, 2024. doi: 10.1093/jtm/taae038.

[21] M. A. Alam, R. Ahmed, and S. Rahman, "Real-Time Crowd Monitoring Using Deep Learning," IEEE Access, 2021. doi: 10.1109/ACCESS.2021.3052880.

[22] H. Fradi, A. Rehman, and M. Bennamoun, “Crowd Density Estimation Using Deep Learning Techniques,” Pattern Recognition Letters, 2020. doi: 10.1016/j.patrec.2020.05.023.

[23] R. K. Gupta and S. Sharma, “AI-Based Early Warning Systems in Smart Cities,” Future Internet, vol. 16, 2024. doi: 10.3390/fi16010018.

[24] A. Alotaibi, M. Alharbi, and N. Khan, “IoT and AI-Based Smart Hajj Monitoring System,” Sensors, 2023. doi: 10.3390/s23125601.

[25] N. Ahmed, S. Paul, and T. Kim, “Context-Aware Intelligent Systems for Smart Environments,” Future Generation Computer Systems, 2022. doi: 10.1016/j.future.2022.04.012.

[26] UNESCO, Recommendation on the Ethics of Artificial Intelligence, Paris, France, 2021. [Online]. Available: https://unesdoc.unesco.org/ark:/48223/pf0000381137

[27] OECD, OECD Principles on Artificial Intelligence, 2019. [Online]. Available: https://oecd.ai/en/ai-principles

[28] European Commission, Ethics Guidelines for Trustworthy Artificial Intelligence, Brussels, Belgium, 2019. [Online]. Available: https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[29] Saudi Data and Artificial Intelligence Authority (SDAIA), National Strategy for Data and Artificial Intelligence, Saudi Arabia. [Online]. Available: https://www.sdaia.gov.sa

[30] World Health Organization, Public Health for Mass Gatherings: Key Considerations, Geneva, Switzerland. [Online]. Available: https://www.who.int